AKTU EXAM READY

Computer Networks
Unit 3

Network Layer - Complete Summary

Topic: IPv4, IPv6, Addressing, Routing, ICMP, Congestion

Format: Point-wise | Easy Language

Target: AKTU Semester Examination

Quick Revision Notes for Maximum Marks

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1. Network Layer Basics

What is Network Layer?

Network Layer is the **3rd layer** of the OSI model. It is responsible for **host-to-host delivery** of packets.

Key Responsibilities

- **Logical Addressing:** Assigns IP addresses to identify source and destination.
- **Routing:** Decides the best path to send packets from source to destination through routers.
- **Packet Creation:** Creates packets from transport layer data; adds logical addresses in header.
- **Fragmentation:** Divides large packets into smaller fragments if needed.

Important Points

- Network layer uses **datagram approach** (connectionless).
- Each packet is handled **independently** and may follow different routes.
- Packets may arrive **out of order**, get **lost**, or **corrupted**.

EXAM TIP: AKTU frequently asks to "Define delivery in Network Layer" and "Discuss the role of logical addressing."

2. IPv4 Protocol

IPv4 (Internet Protocol version 4)

- IPv4 is an **unreliable, connectionless** datagram protocol.
- Provides **no error control** or **flow control** (only error detection on header).
- For reliability, it must be paired with **TCP**.
- Address size: **32 bits** (4 bytes).
- Address space: **2^32 = 4,294,967,296** addresses.

2.1 IPv4 Datagram Format (Header Fields)

A datagram has two parts: **Header (20 to 60 bytes)** + **Data**. Below are the header fields:

Field	Size	Description
Version (VER)	4 bits	IP version (4 for IPv4, 6 for IPv6)
Header Length (HLEN)	4 bits	Header length in 4-byte words. Default = 5 (5 x 4 = 20 bytes). Max = 15 (60 bytes)
Total Length	16 bits	Total size of datagram (header + data). Max = 65,535 bytes
Service Type	8 bits	Defines priority (precedence bits) or differentiated services
Identification	16 bits	Unique ID for each datagram; copied to all fragments
Flags	3 bits	Bit 1: Reserved; Bit 2: Do Not Fragment; Bit 3: More Fragments
Fragmentation Offset	13 bits	Position of fragment in original datagram (measured in 8-byte units)
Time to Live (TTL)	8 bits	Max number of hops (routers) a packet can visit. Decremented by 1 at each router. If 0, packet is discarded.
Protocol	8 bits	Identifies upper-layer protocol (TCP=6, UDP=17, ICMP=1, IGMP=2)

Checksum	16 bits	Error detection for header only
Source Address	32 bits	IP address of sender (remains unchanged)
Destination Address	32 bits	IP address of receiver (remains unchanged)

Quick Recall - IPv4 Header

Fixed header = 20 bytes | **Max header = 60 bytes** (40 bytes options) | **Max datagram size = 65,535 bytes**

Service Type Bits (D, T, R, C)

- **D** = Minimize Delay
- **T** = Maximize Throughput
- **R** = Maximize Reliability
- **C** = Minimize Cost

2.2 Fragmentation

What is Fragmentation?

When a datagram travels through different networks, each network has a different **MTU (Maximum Transmission Unit)**. If the datagram is larger than the MTU, it must be divided into smaller pieces called **fragments**.

Fields Related to Fragmentation

- **Identification:** All fragments of the same datagram carry the same ID value.
- **Flags (3 bits):**
  - Bit 1: Reserved (always 0)
  - Bit 2: **Do Not Fragment** (1 = don't fragment, 0 = can fragment)
  - Bit 3: **More Fragments** (1 = more fragments coming, 0 = last fragment)
- **Fragmentation Offset:** Shows the relative position of the fragment data in the original datagram. Measured in **units of 8 bytes**.

IP Header Options

Option	Purpose

Record Route	Records routers that handled the datagram
Strict Source Route	Predetermines exact route for the datagram
Loose Source Route	Predetermines route but allows extra routers
Timestamp	Records time of processing at each router

EXAM TIP: AKTU asks: "Define IPv4 header Format in detail" - Draw the header diagram and explain all fields.

3. IPv4 Addressing

IPv4 Address

- A **32-bit address** that uniquely identifies a device on the Internet.
- Two devices on the Internet can never have the same IP address at the same time.
- If a device has m connections to the Internet, it needs m IP addresses.
- Two notations: **Binary** (32 bits) and **Dotted-Decimal** (e.g., 192.168.1.1)

3.1 Classful Addressing

In classful addressing, the address space is divided into **5 classes: A, B, C, D, and E**.

Class	First Bits	Range	Network Bits	Host Bits	Use
A	0	1.0.0.0 to 126.255.255.255	8	24	Large organizations
B	10	128.0.0.0 to 191.255.255.255	16	16	Midsize organizations
C	110	192.0.0.0 to 223.255.255.255	24	8	Small organizations

D	1110	224.0.0.0 to 239.255.255.255	-	-	Multicast (groups)
E	1111	240.0.0.0 to 255.255.255.255	-	-	Reserved/Experimental

Important Rules

- **Network Address:** First IP of any network (all host bits = 0). Not assignable to hosts.
- **Broadcast Address:** Last IP of any network (all host bits = 1). Not assignable to hosts.
- **Usable Hosts per Network = 2^(host bits) - 2**

Default Masks

Class	Default Mask	CIDR
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

Flaws in Classful Addressing

1. Class A block is **too large** for most organizations (wastage).
2. Class B block is **too large** for many organizations.
3. Class C block is **too small** for many organizations.
4. Class D addresses were **over-provisioned** for multicast.
5. Class E addresses were **reserved and wasted**.

Result: Classful addressing caused massive address wastage and was replaced by **classless addressing**.

3.2 Subnetting

What is Subnetting?

Subnetting is the process of **dividing a large network into smaller sub-networks (subnets)**. Each department in a company can have its own subnet for better management, security, and performance.

Subnetting Rules

- Work only with the **host portion** of the IP address.
- Convert some **leftmost host bits into subnet bits**.
- Do NOT touch the original network portion.

Subnetting Formulas

Remember These Formulas

- **Number of Subnets = 2^(subnet bits)**
- **Number of Hosts per Subnet = 2^(host bits) - 2**
- **Subnet Mask = Original Mask + Subnet Bits**

Example: Divide 200.1.2.0 into 5 subnets

- Given: 200.1.2.0 is Class C, default mask = 255.255.255.0 (/24)
- For 5 subnets, we need at least **3 subnet bits** (2^2 = 4 < 5, 2^3 = 8 >= 5)
- New subnet mask = /24 + 3 = **/27** = 255.255.255.224
- Each subnet has 2^(5) - 2 = **30 usable hosts**
- Subnet 1: 200.1.2.0 to 200.1.2.31
- Subnet 2: 200.1.2.32 to 200.1.2.63
- Subnet 3: 200.1.2.64 to 200.1.2.95
- Subnet 4: 200.1.2.96 to 200.1.2.127
- Subnet 5: 200.1.2.128 to 200.1.2.159

EXAM TIP: AKTU frequently asks numerical questions on subnetting. Practice finding subnet ranges and number of hosts.

3.3 Classless Addressing (CIDR)

Classless Inter-Domain Routing (CIDR)

- Designed to overcome **address depletion** and provide more efficient allocation.
- IP addresses are allocated based on **network prefix**, not by class.
- Uses **slash notation**: x.y.z.t/n where n = number of network bits.
- Mask: n leftmost bits are 1s, remaining (32-n) bits are 0s.

Restrictions on Address Blocks

1. Addresses in a block must be **contiguous**.
2. Number of addresses must be a **power of 2** (1, 2, 4, 8, ...).
3. First address must be **evenly divisible** by the number of addresses.

Calculating Block Properties

For a given address **x.y.z.t/n**:

- **First Address:** Set (32-n) rightmost bits to 0.
- **Last Address:** Set (32-n) rightmost bits to 1.
- **Number of Addresses:** 2^(32-n).

Example: 205.16.37.39/28

- Mask: /28 means first 28 bits are network, last 4 bits are host.
- First address: 205.16.37.32 (set last 4 bits to 0000)
- Last address: 205.16.37.47 (set last 4 bits to 1111)
- Number of addresses: 2^4 = **16 addresses**

4. IPv6 Protocol

Why IPv6?

- **Address depletion:** IPv4's 4 billion addresses were running out.
- **Real-time support:** Better support for audio/video transmission.
- **Security:** Built-in encryption and authentication.
- IPv6 address is **128 bits** long (16 bytes).
- Address space: **2^128** (approximately 3.4 x 10^38).
- Uses **Hexadecimal Colon Notation**: 8 groups of 4 hex digits separated by colons.

4.1 IPv6 Header Format

Field	Size	Description
Version	4 bits	Value = 6 for IPv6
Priority	4 bits	Packet priority during congestion
Flow Label	20 bits	Identifies a flow of packets needing special handling
Payload Length	16 bits	Length of IP datagram excluding base header
Next Header	8 bits	Defines header that follows (similar to Protocol field in IPv4)
Hop Limit	8 bits	Same as TTL in IPv4
Source Address	128 bits	IP address of sender
Destination Address	128 bits	IP address of receiver

Quick Recall - IPv6 Header

Fixed header size = 40 bytes (no options in base header) | Extension headers used for optional features.

Extension Headers in IPv6

Code	Next Header
0	Hop-by-hop options
6	TCP
17	UDP
43	Source routing
44	Fragmentation
50	Encrypted security payload
51	Authentication
59	Null (no next header)
60	Destination options

4.2 IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Address Space	~4.3 x 10^9	~3.4 x 10^38
Header Size	20-60 bytes (variable)	40 bytes (fixed)
Representation	Decimal (dotted)	Hexadecimal (colon)
Checksum	Present in header	Removed (handled by upper layers)
Fragmentation	By sender and routers	Only by sender
Security (IPSec)	Optional	Inbuilt/Mandatory
Flow Label	Not available	Available
Classes	Classful (A, B, C, D, E)	No classes

Encryption	Not provided	Provided
Authentication	Not provided	Provided

Advantages of IPv6 over IPv4

1. **Larger Address Space:** 128-bit addresses provide virtually unlimited addresses.
2. **Better Header Format:** Options separated from base header, speeding up routing.
3. **New Options:** Allows additional functionalities.
4. **Allowance for Extension:** Protocol can be extended for new technologies.
5. **Resource Allocation:** Flow label enables special handling for real-time audio/video.
6. **More Security:** Built-in encryption and authentication.

> **EXAM TIP:** AKTU frequently asks: "Illustrate the difference between IPv4 and IPv6" and "Discuss advantages of IPv6 over IPv4."

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5. Address Mapping Protocols

5.1 ARP (Address Resolution Protocol)

ARP - Logical to Physical Address Mapping

- When a host/router wants to send an IP datagram, it has the **IP address** of the receiver.
- But the datagram must be encapsulated in a **frame** for the physical network.
- So the sender needs the **physical (MAC) address** of the receiver.
- ARP is used to find the MAC address when the IP address is known.

How ARP Works

1. Sender **broadcasts** an ARP query packet containing:
    - Sender's IP and MAC address
    - Target IP address (looking for its MAC)
2. The **intended recipient** recognizes its IP address.

3. Recipient sends an **ARP reply** (unicast) with its MAC address.
4. Sender caches this IP-to-MAC mapping for future use.

RARP (Reverse ARP) - Physical to Logical

- Finds the **IP address** when MAC address is known.
- Used by **diskless workstations** that know their MAC but not IP.
- RARP request is **broadcast** on local network.
- **Problem:** RARP server must be on the same network; doesn't work across routers.

5.2 BOOTP and DHCP

BOOTP (Bootstrap Protocol)

- Application layer protocol for **physical-to-logical address mapping**.
- Uses **static table** filled by admin (MAC to IP mapping).
- Client uses **0.0.0.0** as source and **255.255.255.255** as destination.
- Requires a **relay agent** if client and server are on different networks.
- **Limitations:**
  - Static configuration - cannot handle mobile hosts.
  - Cannot assign temporary IP addresses.

DHCP (Dynamic Host Configuration Protocol)

> **Why DHCP over BOOTP?**
>
> DHCP provides both **static and dynamic** address allocation. It solves BOOTP's problems by assigning IP addresses on demand with a **lease time**.

Types of Address Allocation

1. **Static Allocation:** DHCP checks its static database for a permanent IP matching the client's MAC address.
2. **Dynamic Allocation:**
   - DHCP maintains a **pool of available IP addresses**.
   - If no static entry exists, assigns an IP from the pool for a **negotiable period (lease time)**.
   - When lease expires, client must **renew** or stop using the IP.

3. **Manual Allocation:** Admin manually assigns IP to a specific MAC.

> **EXAM TIP:** AKTU asks: "How is BOOTP different from DHCP?" Focus on static vs dynamic allocation and lease mechanism.

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6. ICMP Protocol

> **ICMP (Internet Control Message Protocol)**
>
> - ICMP is a **companion protocol to IP**.
> - IP is unreliable and has **no error-reporting** mechanism.
> - ICMP provides **error reporting** and **query** capabilities.
> - ICMP messages are encapsulated in **IP datagrams**.
> - ICMP does **NOT correct errors** - it only reports them.

ICMP Message Format

- **8-byte header** + variable-size data section.
- **Type (8 bits):** Defines the type of message.
- **Code (8 bits):** Reason for the particular message type.
- **Checksum (16 bits):** Error checking over the entire message.

Types of ICMP Messages

A. Error Reporting Messages (5 Types)

1. **Destination Unreachable:**
    - Sent when a router cannot route a datagram or host cannot deliver it.
    - Datagram is discarded and message sent back to source.
2. **Source Quench:**
    - Used for **congestion control**.
    - Router discards datagrams and asks source to slow down.
3. **Time Exceeded:**
    - When TTL reaches 0, router discards packet and sends this message.

   - Also sent when all fragments don't arrive within time limit.
4. **Parameter Problem:**
   - Sent when router/host finds an ambiguous or missing value in any header field.
5. **Redirection:**
   - Router tells a host to use a different router for a particular destination.
   - Helps hosts update their routing tables.

B. Query Messages

1. **Echo Request and Reply (Type 8 / 0):**
   - Used by **PING** command to check connectivity.
   - Determines if two systems can communicate.
2. **Timestamp Request and Reply:**
   - Used to determine **round-trip time** between hosts/routers.
3. **Address-Mask Request and Reply:**
   - Host asks router for its subnet mask.
4. **Router Solicitation and Advertisement:**
   - Host broadcasts to discover routers on its network.
   - Routers respond with their routing information.

> **EXAM TIP:** AKTU asks: "Explain ICMP protocol and its application in real-world scenarios" - Discuss all 5 error types and query messages with PING as example.

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7. Routing Protocols

> **What is Routing?**
>
> Routing is the process of selecting the **best path** for data packets to travel from source to destination across networks. A **routing table** helps routers make this decision.

Types of Routing Tables

- **Static Routing Table:** Routes entered manually by admin. Used in small, stable networks.
- **Dynamic Routing Table:** Updated automatically using routing protocols (RIP, OSPF, BGP). Adapts to network changes.

Autonomous Systems (AS)

- Internet is divided into **Autonomous Systems** - groups of networks under single administration.
- **Intradomain Routing:** Routing inside an AS (RIP, OSPF).
- **Interdomain Routing:** Routing between ASes (BGP).

Protocol	Type	Routing Category
RIP	Distance Vector	Intradomain
OSPF	Link State	Intradomain
BGP	Path Vector	Interdomain

7.1 Distance Vector Routing (DVR)

> **DVR Concept**
>
> Each node maintains a **vector (table)** of minimum distances to every other node. The table shows the **next hop** to reach each destination. Uses **RIP protocol** in practice.

DVR Algorithm - Three Steps

Step 1: Initialization

- Each node knows only the distance to its **immediate neighbors**.
- Distance to non-neighbors is marked as **infinity (unreachable)**.

Step 2: Sharing

- Each node sends its routing table (first two columns) to **immediate neighbors** periodically.

- Sharing happens when there is a **change** or at **regular intervals**.

Step 3: Updating

When a node receives a table from a neighbor:

1. Add the **cost to reach the neighbor** to each value received.
2. Add the **neighbor's name as next hop** for each route.
3. Compare with existing table:
    - If **next hop is different**, choose the row with smaller cost (keep old if tied).
    - If **next hop is the same**, choose the new row (updated information).

Advantages of DVR

- Simple to implement.
- Requires less memory than link state.

Disadvantages of DVR

- Slow convergence (takes time to reach stable state).
- Suffers from **Count to Infinity Problem**.
- Creates routing loops.

7.4 Count to Infinity Problem

> **What is Count to Infinity?**
>
> Distance Vector Routing reacts **rapidly to good news** (shorter path found) but **slowly to bad news** (link failure). This causes routers to incrementally increase distance to infinity.

Example Scenario

1. Initially, all routers know paths to A: B=1, C=2, D=3, E=4 hops.
2. Suddenly, link A-B is cut (A goes down).
3. B doesn't hear from A. But C says: "I have a path to A of length 2."
4. B thinks it can reach A via C with path length = 3. (B doesn't know C's path goes through B itself!)
5. On next exchange, C notices neighbors claim path to A of length 3. C updates its distance to 4.

6. This continues until all routers count up to infinity.

Solutions to Count to Infinity

- **Split Horizon:** Don't advertise a route back to the neighbor you learned it from.
- **Poison Reverse:** Advertise the route back with infinite metric.
- **Hold-down Timer:** Don't accept updates for a route for a certain period after a failure.
- **Define a Maximum:** Set a finite number (e.g., 16) as "infinity" to stop counting.

EXAM TIP: AKTU frequently asks: "Describe count to infinity problem with an example" - Explain the scenario with routers A, B, C, D, E step by step.

7.2 Link State Routing

Link State Concept

Each router knows the state (cost, condition) of its own links and shares this information with **all other routers**. Every router builds a complete map of the network topology. Uses **OSPF protocol** in practice.

Four Steps to Build Routing Tables

Step 1: Creation of Link State Packet (LSP)

- Each router creates LSP containing:
  - **Node Identity**
  - **List of Links** (neighbors and costs)
  - **Sequence Number** (to identify new vs old LSPs)
  - **Age** (to remove outdated LSPs)
- LSPs are generated when topology changes AND periodically.

Step 2: Flooding of LSPs

- Each node sends LSP copies out of **all interfaces**.
- Receiving node compares sequence number with its copy.
- If newer: discards old, keeps new, and forwards to all interfaces **except the incoming one**.

- If older: discards the LSP.

Step 3: Formation of Shortest Path Tree

- Each node now has the **complete network topology**.
- Uses **Dijkstra's Algorithm** to find shortest path to all other nodes.

Step 4: Calculation of Routing Table

- Based on the shortest path tree, each router builds its routing table.

Advantages of Link State

1. **Fast convergence** - responds immediately to network changes.
2. **No routing loops** - complete view of network.
3. **Bandwidth efficient** - no periodic full table updates (only LSP flooding).
4. Better for **large and complex networks**.

Disadvantages of Link State

- Requires **more memory and CPU** (stores neighbor table, link state database, AND routing table).
- Initial LSP flooding can cause **high traffic**.

7.3 Path Vector Routing (BGP)

> **BGP (Border Gateway Protocol)**
>
> Path Vector Routing is used for **interdomain routing** (between Autonomous Systems). Each AS has a **speaker node** that represents the entire AS and communicates with other AS speakers.

How Path Vector Works

1. **Initialization:** Each speaker node knows about nodes within its own AS.
2. **Sharing:** Speaker nodes share their tables with **immediate neighboring AS speakers**.
3. **Updating:** When a speaker receives a table, it adds new nodes and prepends its own AS and sender's AS to the path.

Three Policies in Path Vector

1. **Loop Prevention:** If a router sees its own AS in the path, it ignores the message (loop detected).
2. **Policy Routing:** Router can ignore paths that violate its organization's policy.
3. **Optimum Path:** Choose the path with the **smallest number of ASes**.

Hierarchical Routing

- For very large networks, routers are divided into **regions**.
- Each router knows details about its own region but nothing about internal structure of other regions.
- Regions can be grouped into **clusters, zones, and groups**.

> **EXAM TIP:** AKTU asks: "Explain all Interdomain and Intradomain routing algorithms" - Cover DVR (RIP), Link State (OSPF), and Path Vector (BGP) with their categories.

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8. Congestion Control

> **What is Congestion?**
>
> - When **too many packets** are present in the network, it causes packet delay and loss.
> - This degrades network performance and is called **congestion**.
> - Congestion occurs when **load** > **capacity**.
> - Both **network layer** and **transport layer** share responsibility for handling it.

Congestion Control Categories

1. **Open-Loop Congestion Control:** Prevents congestion **before** it happens.
2. **Closed-Loop Congestion Control:** Removes congestion **after** it happens.

8.1 Open-Loop Congestion Control (Prevention)

1. **Retransmission Policy:**

- Design retransmission timer carefully to prevent unnecessary retransmissions.
- Retransmitting too fast increases congestion.

2. **Window Policy:**
   - **Selective Repeat** is better than Go-Back-N for congestion control.
   - Selective Repeat only retransmits lost packets, reducing network load.
3. **Acknowledgment Policy:**
   - Receiver can send fewer acknowledgments to reduce network traffic.
   - Ack only when necessary or after receiving N packets.
4. **Discarding Policy:**
   - Router can discard less sensitive packets when congestion is likely.
5. **Admission Policy:**
   - Router can **deny new virtual circuit connections** if congestion exists or is likely.

8.2 Closed-Loop Congestion Control (Removal)

1. **Backpressure:**
   - A congested node stops receiving data from its **upstream node**.
   - The upstream node also becomes congested and stops receiving from its upstream.
   - Propagates in **opposite direction of data flow** back to the source.
   - This is a **node-to-node** congestion control.
2. **Choke Packet:**
   - A congested router sends a **warning packet** directly to the source.
   - Example: **ICMP Source Quench** message.
   - Source is warned to slow down.
3. **Implicit Signaling:**
   - Source **guesses** congestion from symptoms like delayed ACKs.
   - Example: **TCP congestion control** slows down when ACKs are delayed.
   - No direct communication about congestion.
4. **Explicit Signaling:**

- Congested node **explicitly sends a signal** in data packets.
- **Backward Signaling:** Bit set in packet moving opposite to congestion direction (warns source).
- **Forward Signaling:** Bit set in packet moving toward congestion direction (warns destination).

> **EXAM TIP:** AKTU asks: "What is congestion? Describe techniques that prevent congestion" - Cover both open-loop and closed-loop mechanisms with all sub-types.

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9. AKTU Previous Year Questions

1. Define IPv4 header Format in detail. **(AKTU 2023-24)**
2. Define delivery in Network Layer. **(AKTU 2023-24)**
3. Discuss the role of logical addressing. **(AKTU 2021-22)**
4. Discuss about the IP ranges of Class A, B, C and D. **(AKTU 2021-22)**
5. If a class B network has subnet mask 255.255.248.0, what is the maximum number of hosts per subnet? **(AKTU 2018-19)**
6. What is IP Addressing? How it is classified? How is subnet addressing performed? **(AKTU 2018-19)**
7. The IP network 200.198.160.0 is using subnet mask 255.255.255.224. Draw the subnets. **(AKTU 2022-23, 2018-19)**
8. Divide the network with IP address 200.1.2.0 into 5 subnets. **(AKTU 2021-22, 2018-19)**
9. Illustrate the difference between IPv4 and IPv6. **(AKTU 2022-23)**
10. Discuss advantages of IPv6 over IPv4. **(AKTU 2021-22)**
11. Explain ICMP protocol and its application in real-world scenarios. **(AKTU 2022-23)**
12. Explain ARP. **(AKTU 2022-23)**
13. Discuss in detail about ICMP role in network layer. **(AKTU 2021-22)**
14. How is BOOTP different from DHCP? **(AKTU 2018-19)**
15. Explain all Interdomain and Intradomain routing algorithms. **(AKTU 2023-24)**
16. Describe "count to infinity problem" with an example. **(AKTU 2022-23, 2018-19, 2017-18, 2016-17)**
17. Explain distance vector routing (DVR) with working example in detail. **(AKTU 2022-23)**

18. What is unicast routing? Explain all unicast routing protocols in detail. **(AKTU 2018-19)**
19. Describe few reasons of congestion in the network. **(AKTU 2017-18)**
20. What is congestion? Briefly describe techniques that prevent congestion. **(AKTU 2017-18)**

Most Repeated Questions (High Priority)

1. **Count to Infinity Problem** - Asked in 2022-23, 2018-19, 2017-18, 2016-17
2. **Subnetting Numericals** - Asked almost every year
3. **IPv4 Header Format** - Very frequently asked
4. **ICMP Protocol** - Regularly asked
5. **Routing Algorithms (DVR, Link State)** - Frequently asked

--- All the Best for Your AKTU Exam! ---